

$$f: (a, b) \rightarrow \mathbb{R}$$

$$c \in (a, b)$$

$$u(x) = \frac{f(x) - f(c)}{x - c}$$

$$R(h) = \frac{f(c+h) - f(c)}{h}$$

$$f'(c) = \lim_{x \rightarrow c} u(x) = \lim_{h \rightarrow 0} R(h)$$

$$r(x) = f(c) + f'(c)(x-c) \quad y = f(c) + f'(c)(x-c)$$

$$\lim_{x \rightarrow c} \frac{f(x) - r(x)}{x - c} = 0$$

Regole di derivazione

$$f: (a, b) \rightarrow \mathbb{R}$$

$$m \quad c \in (a, b) \quad k \in \mathbb{R}$$

$$g(x) = k f(x)$$

$$\text{Si ha: } \exists g'(c) = k f'(c)$$

$$\frac{k f(x) - k f(c)}{x - c} = k \quad \frac{f(x) - f(c)}{x - c} \rightarrow k f'(c)$$

$$2) f, g: (a, b) \rightarrow \mathbb{R} \quad m \quad c \in (a, b)$$

$$r(x) = f(x) + g(x) \quad \text{Si ha } r'(c) = f'(c) + g'(c)$$

$$\frac{r(x) - r(c)}{x - c} = \frac{f(x) + g(x) - f(c) - g(c)}{x - c} = \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c}$$

$$\downarrow$$

$$f'(c) \quad g'(c)$$

$$3) f, g: (a, b) \rightarrow \mathbb{R} \quad m \quad c \in (a, b)$$

$$r(x) = f(x) g(x) \quad \text{Si ha } r'(c) = f'(c) g(c) + f(c) g'(c)$$

NO DIM

4)

$$f: (a, b) \rightarrow \mathbb{R} \quad m \quad c \in (a, b) \quad f(c) \neq 0$$

(f derivabile $\Rightarrow f$ cont in c $f(c) \neq 0 \Rightarrow f(x) \neq 0$ in un intorno di c)

$$F(x) = \frac{1}{f(x)} \quad (\text{esiste in un intorno di } c)$$

$$\text{Si ha: } \exists f'(c) = \frac{-f'(c)}{(f(c))^2}$$

No Dim

$$5) f, g: (a, b) \rightarrow \mathbb{R} \quad \text{der. in } c \in (a, b) \quad g(c) \neq 0$$

$$q(x) = \frac{f(x)}{g(x)} \quad \text{Si ha } q'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$$

infatti

$$q(x) = f(x) \frac{1}{g(x)} \Rightarrow q'(c) = f'(c) \frac{1}{g(c)} + f(c) \frac{-g'(c)}{(g(c))^2} = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$$

teorema di derivazione delle funzioni composte

$$\text{IP: } f: (d, \beta) \rightarrow \mathbb{R} \quad g: (a, b) \rightarrow (d, \beta) \quad c \in (a, b) \\ F(x) = f(g(x))$$

$$\exists g'(c), \exists f'(g(c))$$

No Dim

$$\text{TS: } \exists F'(c) = f'(g(c)) g'(c)$$

Teoremi di derivazione delle funzioni inverse

$f: (a, b) \rightarrow (d, \beta)$ stretta (o densa) e continua

$f': (d, \beta) \rightarrow (a, b)$ suriettivo //

Sia $\gamma \in (d, \beta)$ $\gamma = f(c)$, $c \in (a, b)$

$\exists f'(c) \neq 0$

$$\text{ts } \exists (f^{-1})'(\gamma) = \frac{1}{f'(c)}$$

NO DIM

teoremi

Praticamente per trovare $(f^{-1})'(\gamma)$ si deve risolvere l'eq $f(x) = \gamma$ per trovare c , calcolare $f'(c)$ se $\neq 0$ si conclude

Derivate delle funzioni elementari

$$f(x) = k \quad \forall x \in \mathbb{R} \Rightarrow f'(x) = 0 \quad \forall x \in \mathbb{R}$$

$$f(x) = x \quad " \Rightarrow f'(x) = 1$$

$$f(x) = x^2 \quad r(x) = \frac{x^2 - c^2}{x - c} = \frac{(x-c)(x+c)}{(x-c)} \rightarrow 2c$$

$$\Rightarrow f'(x) = 2x \quad \forall x \in \mathbb{R}$$

$$f(x) = x^3 \quad r(x) = \frac{x^3 - c^3}{x - c} = \frac{(x-c)(x^2 + xc + c^2)}{(x-c)} \rightarrow 3c^2$$

$$\Rightarrow f'(x) = 3x^2 \quad \forall x \in \mathbb{R}$$

$\vdots$

$$f(x) = x^n$$

$$f'(x) = nx^{n-1} \quad \forall x \in \mathbb{R}$$

$$\frac{x^n - c^n}{x - c} = \frac{(x - c)}{(x - c)} (x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1}) \rightarrow \sum_{m=0}^{n-1} c^m$$

$$f(x) = x^{-n} = \frac{1}{x^n} \quad f'(x) = \frac{-n x^{n-1}}{x^{2n}} = -n x^{n-1-2n} = -n x^{-n-1} \quad (x \neq 0)$$

$$f(x) = e^x \quad \pi(x) = \frac{e^x - e^c}{x - c} = e^c \frac{e^{x-c} - 1}{x - c} \rightarrow e^c$$

$\downarrow$
 $\frac{1}{1}$

$$f'(x) = e^x \quad \forall x \in \mathbb{R}$$

$$\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = \log e$$

$$f(x) = a^x \quad (a > 0, a \neq 1)$$

$$f'(x) = a^x \log a$$

$$\lim_{t \rightarrow 0} \frac{\log(t+1)}{t} = \frac{1}{1}$$

$$f(x) = \log x \quad c > 0 \quad \pi(x) = \frac{\log x - \log c}{x - c} = \frac{\log \frac{x}{c} - 1 + 1}{c \left(\frac{x}{c} - 1 \right)}$$

$\downarrow$
 $\frac{1}{1}$

$$f'(x) = \frac{1}{x} \quad \forall x > 0$$

possiamo vederla anche come funzione inversa

$$f(x) = e^x :]-\infty; +\infty[\rightarrow]0; +\infty[$$

$$f^{-1}(y) = \log y :]0; +\infty[\rightarrow]-\infty; +\infty[$$

$$y \in]0; +\infty[$$

$$c: f(c) = y \quad e^c = y \Rightarrow c = \log y$$

$$f'(c) = \log y \neq 0$$

$$(f^{-1})'(y) = \frac{1}{\log y} = \frac{1}{c}$$

Se $e \neq c$ $f(x) = \log_e x \Rightarrow f'(x) = \frac{1}{x} \log_e e$

$f(x) = (x) \neq f'(0)$ me $f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases} = \frac{|x|}{x}$

("segno di x")

Se f deriv. e $\neq 0$ e con $f(x) = |f(x)|$

$F' = \frac{|f(x)|}{f(x)} f'(x)$

Consideriamo ora $f(x) = \log |f(x)|$ $f'(x) = \frac{1}{|f(x)|} \frac{|f(x)|}{f(x)}$
 $\downarrow$
 $f'(x) = \frac{f'(x)}{f(x)}$

es: $F(x) = \log |x^3 - 3x|$ in tutti i punti in cui $x^3 - 3x \neq 0$ si ha

$f'(x) = \frac{3x^2 - 3}{x^2 - 3x}$

$f(x) = x^d$ (d non intero) def in $[0, +\infty[$ e $d > 0$
in $]0, +\infty[$ e $d < 0$

$C > 0$ + > 0 + = e^{log+}

$x^d = e^{\log x^d} = e^{d \log x}$

$f'(x) = e^{d \log x} \cdot d \cdot \frac{1}{x} = x^d \cdot \frac{1}{x} = d x^{d-1}$

es: $d = \frac{1}{2}$ $f(x) = \sqrt{x}$ $f'(x) = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2\sqrt{x}}$ $x > 0$

se $l < 1$ $\nexists f'(0)$ $\frac{x^l - e^l}{x - e}$ diverge per $x \rightarrow 0$

ma se di $f'(c)$ si può scrivere $(D(f(x)))_{x=c}$

$$\text{es: } D(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\text{con } f(x) = x^n \quad n \in \mathbb{N}$$

$$f'(x) = nx^{n-1}$$

$$f''(x) = n(n-1)x^{n-2}$$

$$f'''(x) = n(n-1)(n-2)x^{n-3}$$

$$\vdots$$
$$f^{(n)}(x) = n(n-1)(n-2) \dots 1 \cdot x^0 = n!$$

$$f^{(k)}(x) = 0 \quad \forall k > n$$

Funzioni trigonometriche

$$f(x) = \cos x \quad f(h) = \frac{\cos(c+h) - \cos c}{h} =$$
$$= \frac{\cos c \cosh - \sin c \sinh - \cos c}{h} = \cos c \frac{\cosh - 1}{h} -$$

$$\sin c \frac{\sinh}{h} \rightarrow -\sin c$$

$$f(x) = \cos x: [0, \pi] \rightarrow [-1, 1]$$

$$f^{-1}(x) = \arccos [-1, 1] \rightarrow [0, \pi]$$

$$x \in [-1, 1] \quad x = \cos c \quad c \in [0, \pi] \quad f'(c) = -\sin c \neq 0$$

$$\text{per } c \in]0, \pi[$$

$$\Downarrow$$
$$x \in]-1, 1[$$

$$(f^{-1})'(x) = \frac{1}{-\sin c} = - \frac{1}{\sin c} = - \frac{1}{\sqrt{1 - \cos^2 c}} =$$

perché $\sin^2 c = 1 - \cos^2 c$ e $\sin c > 0$

$$= - \frac{1}{\sqrt{1 - \cos^2(\arccos y)}} = \frac{1}{\sqrt{1 - y^2}} \quad \forall y \in]-1, 1[$$

$$f(x) = \sin x \quad h(h) = \frac{\sin(c+h) - \sin c}{h} = \frac{\sin c \cosh + \cos c \sin h - \sin c}{h} = \sin c \frac{\cosh - 1}{h} +$$

$$\cos c \frac{\sin h}{h} \rightarrow \cos c \quad f'(x) = \cos x \quad \forall x$$

$$f(x) = \sin x: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

$$f'(x) = \cos x \quad \sin x: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad y \in [-1, 1]$$

$$c \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]: \sin c = y \quad c = \arcsin y$$

$$f'(c) = \cos c \neq 0 \quad \text{per } c \neq \pm \frac{\pi}{2} \Leftrightarrow y \neq \pm 1$$

$$(f^{-1})'(y) = \frac{1}{\cos c} = \frac{1}{\sqrt{1 - \sin^2 c}} = \frac{1}{\sqrt{1 - \sin^2(\arcsin y)}} = \frac{1}{\sqrt{1 - y^2}}$$

$$f(x) = \tan x = \frac{\sin x}{\cos x} \quad x \neq \frac{\pi}{2} + k\pi$$

$$f'(x) = \frac{\cos x \cos x - \sin(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = 1 + \tan^2 x$$

$$f(x) = \tan x: \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[\rightarrow]-\infty, +\infty[$$

$$f^{-1}(y) = \arctan y:]-\infty, +\infty[\rightarrow \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$$

$$x \in]-\infty; +\infty[\quad y = \tan e \quad c \in]-\frac{\pi}{2}, \frac{\pi}{2}[$$

$$f'(c) = 1 + \tanh^2 c$$

$$(f^{-1})'(x) = \frac{1}{1 + \tanh^2 c} = \frac{1}{1 + \tanh^2(\arctan x)} = \frac{1}{1 + x^2}$$

Exercice:

$$f(x) = 3x^4 - \cos x + e^x$$

$$f'(x) = 12x^3 + \sin x + e^x$$

$$f(x) = (\sin x) \sqrt{x}$$

$$f'(x) = (\cos x) \sqrt{x} + \sin x \cdot \frac{1}{2\sqrt{x}}$$

$$f(x) = \frac{\tan x}{x^6}$$

$$f'(x) = \frac{(1 + \tanh^2 x) x^6 - \tanh x (6x^5)}{x^{12}}$$

$$f(x) = \frac{\arcsin x}{\arccos x}$$

$$f'(x) = \frac{(\arcsin x + \frac{x}{\sqrt{1-x^2}}) \arccos x + \frac{1}{\sqrt{1-x^2}} \times \arcsin x}{\arccos^2 x}$$

$$f(x) = \sin(4x^2 - 1)$$

$$f'(x) = \cos(4x^2 - 1) \cdot 8x$$

$$f(x) = e^{\sqrt{x}}$$

$$f'(x) = e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$$

$$f(x) = \arctg \frac{2x}{x^2-1}$$

$$f'(x) = \frac{1}{1 + \frac{4x^2}{(x^2-1)^2}} \cdot \frac{2(x^2-1) - 2x \cdot 2x}{(x^2-1)^2}$$

$$f(x) = \sqrt[3]{x - \sqrt{\sin x}}$$

$$f'(x) = \frac{1}{3} (x - \sqrt{\sin x})^{\frac{1}{3}-1} \left(1 - \frac{\cos x}{\sqrt{\sin x}} \right)$$

$$f(x) = e^{\arctan x}$$

$$f'(x) = e^{\arctan x} \cdot \frac{1}{1+x^2}$$

$$f(x) = e^{\sin \frac{1}{x}}$$

$$f'(x) = e^{\sin \frac{1}{x}} \cdot \left(\cos \frac{1}{x} \right) \left(\frac{-1}{x^2} \right)$$

$$f(x) = \cos \sqrt{x^2 - \frac{1}{x}}$$

$$f'(x) = -\sin \sqrt{x^2 - \frac{1}{x}} \cdot \frac{1}{2\sqrt{x^2 - \frac{1}{x}}} \cdot \left(2x + \frac{1}{x^2} \right)$$

$$f(x) = \arctan \sqrt{\frac{1+x}{1-x}}$$

$$f'(x) = \frac{1}{1 + \frac{1+x}{1-x}} \cdot \frac{1}{2\sqrt{\frac{1+x}{1-x}}} \cdot \frac{1(1-x) \cdot (-1)(1+x)}{(1-x)^2}$$

$$f(x) = \log 2x$$

$$f'(x) = \frac{1}{2x} \cdot 2 = \frac{2}{2x}$$

$$f(x) = \log \sqrt{\frac{x}{x-1}}$$

$$f'(x) = \frac{1}{\sqrt{\frac{x}{x-1}}} \cdot \frac{1}{2\sqrt{\frac{x}{x-1}}} \cdot \frac{1(x-1) - x \cdot 1}{x^2 + 1 - 2x}$$

$$f(x) = \log |x^4 + 3x|$$

$$D(\log |f|) = \frac{f'}{f}$$

$$f'(x) = \frac{4x^3 + 3}{x^4 + 3x}$$

$$f(x) = \log |\sqrt[4]{x} - x^3|$$

$$f'(x) = \frac{\frac{1}{4} \cdot \frac{1}{\sqrt[4]{x^3}} - 3x^2}{\sqrt[4]{x} - x^3}$$

$$f(x) = \frac{\log \frac{3x}{x-1}}{\arcsin x^2}$$

$$f'(x) = \frac{\frac{x-1}{3x} \cdot \frac{3(x-1) - 3x}{(x-1)^2} \cdot \arcsin x^2 - \log \frac{3x}{x-1} \cdot \frac{1}{\sqrt{1-x^2}} \cdot 2x}{\arcsin^2 x^2}$$

Impostare le regole di derivazione

$$f(x) = \sin(\sin x)$$

$$f'(x) = \cos(\sin x) \cdot \cos x$$

$$f(x) = \cos(\cos(\cos x))$$

$$f'(x) = -\sin(\cos(\cos x)) \cdot (-\sin(\cos x)) \cdot (-\sin x)$$

$$f(x) = x^x = e^{\log x^x} = e^{x \log x}$$

$$f'(x) = e^{x \log x} \cdot \left[1 \cdot \log x + x \cdot \frac{1}{x} \right] = x^x (\log x + 1)$$

$$f(x) = (\sin x)^{\sin x} = e^{\sin x \log \sin x}$$

$$f'(x) = \sin x^{\sin x} \left[\cos x \log(\sin x) + \sin x \frac{\cos x}{\sin x} \right]$$

Per cose

$$f(x) = \log \frac{1}{\tan x}$$

$$f(x) = \sqrt{\cos^3 x - \sin x}$$

$$f(x) = e^{\frac{1}{\arctan x}}$$

$$f(x) = \log \left| \sqrt[5]{x^2 - x} \right|$$

$$f(x) = \tan(\log(\cos x))$$

$$f(x) = \sqrt{x^2 - 1} \sqrt[3]{3x^4 - 2}$$

Applicazioni del calcolo differenziale allo studio delle funzioni

Criterio di monotonia globale

Se $f'(c) > 0$ allora f è cresc. in c

infatti $f'(c) > 0 \Rightarrow \exists(x) > 0$ in un intorno di c

$\Rightarrow f$ cresc. in c

Criterio di strette monotomie

Se $f'(x) > 0 \quad \forall x \in (a, b) \Rightarrow f$ è strett. cresc. in (a, b)

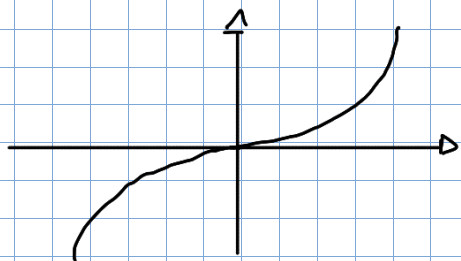
infatti $\exists$ cresc. in ogni punto per il teorema prec.

Il viceversa non vale, nemmeno nelle monotomie locali

f cresc. in $c \Rightarrow f'(c) \geq 0$ (se $f'(c) < 0$ f sarebbe decresc.)

consideriamo $f(x) = x^3$ $f'(x) = 3x^2$

$f'(0) = 0$ ma f è crescente strett



Ci serve allora un criterio di strette monotonia
in cui f' può anche essere 0. Per ovviare dobbiamo
studiare, teoremi fondamentali sul calcolo
differenziale

Teoremi di Fermat

$f: (a, b) \rightarrow \mathbb{R}$ $c \in]a, b[$ punto di estremo relativo

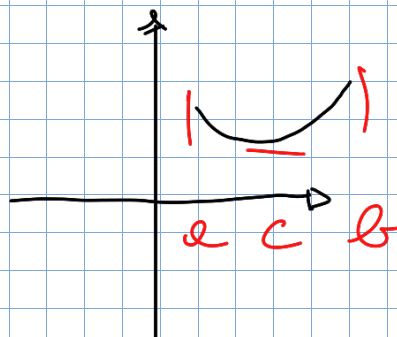
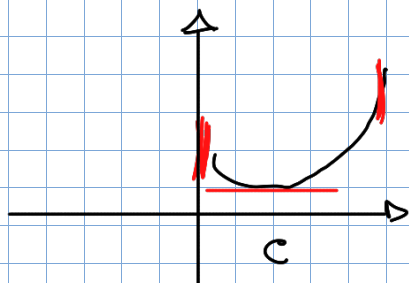
$$\exists f'(c)$$

$$f'(c) = 0$$

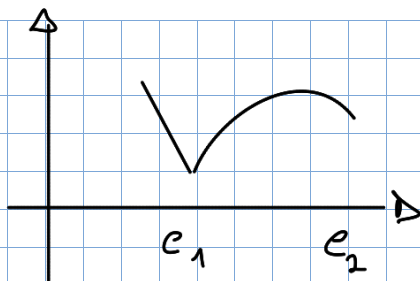
Osservazioni:

1) non vale il viceversa $f(x) = x^3$, $c = 0$

2) non vale se $c = a$ oppure $c = b$



3) Cercheremo i punti di estremo rel fra i punti interni in
cui la derivata è zero (punti stazionari) e quelli
in cui non c'è derivata



C_2 punti stazionari